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MIL-STD-1839C
27 November 2000
SUPERSEDING
MIL-STD-1839B
20 AUGUST 1995

DEPARTMENT OF DEFENSE

STANDARD PRACTICE FOR

CALIBRATION AND MEASUREMENT

REQUIREMENTS

AMSC F4562 AREA QCIC

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FOREWORD

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to AFMETCAL/MLSR, 813 Irving-Wick Dr W, Suite 4M, Heath OH 43056-6116, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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CONTENTS

PARAGRAPH		PAGE
	FOREWORD	ii
1.	SCOPE	1
1.1	Purpose	1
1.2	Applicability	1
1.2.1	Application guidance	1
2.	APPLICABLE DOCUMENTS	1
2.1	General	1
2.2	Government documents	1
2.2.1	Specifications, standards and handbooks	1
2.2.2	Other government documents, drawings, and publications	2
2.3	Order of precedence	2
3.	DEFINITIONS	2
3.1	Calibration	2
3.2	Support Equipment (SE)	3
3.3	Test, Measurement, and Diagnostic Equipment (TMDE)	3
3.4	Measurement traceability	3
3.5	Test Uncertainty Ratio (TUR)	3
3.6	Logistic Support Analysis (LSA)	3
3.7	Design activity	3
3.8	Calibration and Measurement Requirements Summary (CMRS)	3
3.9	Automatic Test Equipment (ATE) Unit Under Test (UUT) interface	3
3.10	Built-in Test (BIT)	4
3.11	Built-in-Test Equipment (BITE)	4
3.12	Calibration interval	4
4.	GENERAL REQUIREMENTS	4
4.1	General measurement requirements	4
5.	DETAILED REQUIREMENTS	4
5.1	System, subsystem, and equipment measurement parameters	4
5.1.1	Assessment of parameters	4
5.1.1.1	Sequence of relationship	4
5.1.1.2	Selecting TMDE	5
5.1.1.3	Description of items	5
5.2	Built-In Test and Built-In Test Equipment	5
5.3	System, subsystem, and equipment test points	5

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CONTENTS

PARAGRAPH		PAGE
5.4	Test, measurement, and diagnostic equipment..	6
5.4.1	Recommended TMDE	6
5.4.2	TMDE in support of TMDE.	6
5.4.3	Test Uncertainty Ratio	6
5.4.4	Automatic test equipment	7
5.4.4.1	ATE design.	7
5.4.4.2	ATE technical description.	7
5.4.4.3	ATE calibration	7
6.	NOTES ..	7
6.1	Intended use	7
6.2	Issue of DoDISS.	7
6.3	Associated Data Item Descriptions (DIDs)	8
6.4	Subject term (keyword) listing.	8
6.5	Changes from previous issue.	8
CONCLUDING MATERIAL.		8

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1. SCOPE

1.1 Purpose. This standard describes the requirement to provide a means for calibration and measurement traceability of all system, subsystem, and equipment parameters to ensure system and equipment operational integrity and accuracy. It defines the requirement for establishing measurement traceability from actual system and equipment level measurements to the National Institute of Standards and Technology (NIST) or other approved measurement sources. These measurements are traced through properly selected and calibrated Test, Measurement, and Diagnostic Equipment (TMDE) or appropriate measurement sensors, through individual Military Department or commercial calibration facilities, to approved national measurement standards.

1.2 Applicability. The requirements of this standard apply to all systems, subsystems, and equipment that require measurements of any type to ensure proper operation.

1.2.1 Application guidance.

a. When Logistics Support Analysis (LSA) is required on an acquisition program, the requirements of this standard should be an integral part of the LSA effort. Data developed as a result of this standard should be documented in and become part of the LSA documentation.

b. When LSA is exempted or the LSA is tailored to exempt development of the data requirements, this standard can become an independent compliance document of the acquisition program and tailored in accordance with the contract.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

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HANDBOOK

DEPARTMENT OF DEFENSE

MIL-HDBK-300 Military Handbook, Technical Information File of Support Equipment

(Unless otherwise indicated, copies of above military handbook is available from the Defense Printing Service, Detachment Office, Philadelphia, ATTN: D-4 Monitor, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAIR 17-35MTL-1	Metrology Requirements List (METRL)
NAVAIR 17-35NCE-1	Navy Calibration Equipment List (NCE)
TB 43-180	Calibration and Repair Requirements for the Maintenance of Army Material
TM-10510-14/1	U.S. Marine Corps Electronics Test Equipment Listing
TO 33K-1-100-1	Technical Manual TMDE Calibration Notes, Maintenance Data Collection Codes, and Calibration Measurement Summaries.
TO 33K-1-100-2 -	Technical Manual TMDE Calibration Interval Technical Order and Work Unit Code Reference Guide
TO 33K-1-101 -	Technical Manual, Calibration Standards and Associated Equipment

(Copies of Air Force TO 33K-1-100-1, 33K-1-100-2, Army TB 43-180, NAVAIR 17-35MTL-1, TO 33K-1-101, or NAVAIR 17-35NCE-1 should be obtained as directed by the contracting officer. The Marine Corps Technical Manual is available from HQMC, Technical Manual Sec., Washington, DC 20380-0001.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence. Nothing in this standard, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Calibration. The set of operations that establish, under specified conditions, the relationship between values of quantities indicated by a measuring instrument or measuring system, or values represented by a material measure or a reference material, and the corresponding values realized by standards.

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3.2 Support Equipment (SE). All equipment used in calibration and maintenance support of mission and operational equipment.

3.3 Test, Measurement, and Diagnostic Equipment (TMDE). Any system or device used to test, measure, evaluate, inspect, or otherwise examine materials, supplies, equipment, or a system to identify and/or isolate any actual or potential malfunction, or to determine compliance with specifications established in technical documents (e.g. research, development, test, and evaluation documents; specifications; engineering drawings; technical orders). TMDE is SE that provides for measurement traceability.

3.4 Measurement traceability. The property of the result of a measurement or the value of a standard whereby it can be related to stated references, usually national or international standards, through an unbroken chain of comparisons all having stated uncertainties. Traceability, as used in this standard, applies to measurements/calibrations made from the prime system or subsystem through an unbroken chain of comparisons to the national reference standards.

3.5 Test Uncertainty Ratio (TUR). The total uncertainty of the unit to be measured or calibrated divided by the total uncertainty of the measuring or generating device used to perform the measurement. For example, if it is required that a system or equipment output parameter's uncertainty is 8% and the uncertainty of the measuring device used to measure the output parameter is 2%, then the TUR is 8 to 2 or 4 to 1.

3.6 Logistics Support Analysis (LSA). The selective application of scientific and engineering efforts undertaken during the acquisition process, as part of the system engineering and design process, to assist in complying with supportability and other Integrated Logistics Support (ILS) objectives. The LSA process is a planned series of tasks performed to examine all elements of a proposed system to determine the logistics support required to keep that system useable for its intended purpose; and to influence the design so both the system and support can be provided at an affordable cost.

3.7 Design activity. A design activity is an activity that has responsibility for the design or modification of an item. The activity may be a U.S. Government, commercial, or nonprofit organization.

3.8 Calibration and Measurement Requirements Summary (CMRS). A report which details the measurement requirements of a system, subsystem, or equipment; the TMDE, and the calibration standards and equipment required to assure traceability of all measurements through the applicable individual military department's metrology and calibration program.

3.9 Automatic Test Equipment (ATE) Unit Under Test (UUT) interface. The point located on the ATE where all input and output signals are accessible for connection to an external UUT or calibration standard.

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3.10 Built-in-Test (BIT). A test approach using built-in test equipment or self-test hardware and software that is internally designed into the supported system, subsystem, or equipment to test all or a part of that system, subsystem, or equipment.

3.11 Built-in-Test Equipment (BITE). Any device which is a part of a system, subsystem, or equipment and is used for the express purpose of testing the system, subsystem or equipment. BITE is an identifiable unit of the system, subsystem, or equipment.

3.12 Calibration interval. The maximum time an item, requiring calibration, may go between calibrations and still achieve its target measurement reliability.

4. GENERAL REQUIREMENTS

4.1 General measurement requirements. The design activity shall ensure the following actions are accomplished for the affected systems and equipment throughout their life cycle.

a. All parameters that require measurement to ensure proper and accurate operation of the system and equipment are identified and documented.

b. Calibration and measurement actions are conducted as required to maintain accurate operation. Such actions are to be accomplished at all applicable system and subsystem levels and are appropriately integrated into the total system requirements.

c. All parameters are readily accessible and measurable in a manner that minimizes the number and duration of the tests required.

d. Measurement traceability of all system and subsystem parameters is documented.

5. DETAILED REQUIREMENTS

5.1 System, subsystem, and equipment measurement parameters.

5.1.1 Assessment of parameters. The design activity shall assess all system, subsystem, and equipment parameters that require measurement or testing to ensure proper system or equipment operation and accuracy, and to ensure intended mission goals are met.

5.1.1.1 Sequence of relationship. This assessment of parameters shall portray the logical sequence of relationship within and between the system, subsystem, and equipment, and identify the parameters requiring measurement and verification.

a. System parameters require measurement and verification to ensure proper operation of the system so the mission requirements of the system can be accomplished.

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b. Subsystem parameters require test and measurement to ensure interchangeability and proper operation of the subsystem when integrated into the complete system.

c. Equipment parameters require test and measurement to ensure interchangeability and proper operation of the equipment when used as part of or connected with the system or subsystem.

5.1.1.2 Selecting TMDE. This assessment of parameters shall be used by the design activity during the evaluation process for recommending or selecting TMDE and developing support documentation.

5.1.1.3 Description of items. The system, subsystem, or equipment, that has parameters that require measurement shall be identified by nomenclature, manufacturer and manufacturer's code, part or model number, and type designation. If an item does not have an approved part number, but does have an approved or proposed end item specification number or system number, that number shall be referenced. Each requirement shall identify:

a. Function: The function (specific input, output, or other characteristic which has units of measurement such as volts, frequency, power, current, length, force, etc.) which will be measured, tested, checked, or adjusted to determine or maintain the item's operational condition.

b. Operational range or specific value: The range of values or actual value that shall be measured to satisfy operational requirements.

c. Operational Tolerance: The tolerance of the range or specific value within which the equipment is required to perform to meet operational specifications.

d. Interval: The design activity shall recommend a maximum time lapse between tests or other method of scheduling tests.

5.2 Built-In Test and Built-In Test Equipment. For parameters of BIT and BITE, or other internal measurements which are part of the operational equipment requiring test or measurement, the nomenclature, manufacturer or code, part number or model number, range and accuracy of the operational equipment shall be identified, as well as the parameters being monitored or generated. When built-in references are employed, a method of test or measurement shall be identified or, if not required, a narrative justification shall be documented.

5.3 System, subsystem, and equipment test points. The design activity shall ensure measurements can be accomplished, test points are identified in technical data, and can be found in the system and/or equipment, and are accessible with minimum disturbance to configuration of the system and/or equipment.

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5.4 Test, Measurement, and Diagnostic Equipment.

5.4.1 Recommended TMDE. The design activity shall ensure that TMDE is recommended to satisfy all measurement requirements identified in accordance with sections 5.1 and 5.2 of this standard.

a. The TMDE recommended shall be capable of functioning in the system operational measurement environment and satisfy all parameters of each measurement (see 5.4.3).

b. The design activity shall assess TMDE specifications in support of measurement requirements identified in accordance with sections 5.1 and 5.2 of this standard. Where several items of TMDE are used in combination, the overall test configuration specification and uncertainty shall be documented.

5.4.2 TMDE in support of TMDE.

a. The design activity shall identify calibration equipment and standards required to support TMDE recommendations which are not currently supportable (i.e. not listed with calibration procedures and intervals) by the applicable military department metrology and calibration program. Refer to Air Force TO 33K-1-100-1, -2, Army TB 43-180, Navy NAVAIR 17-35MTL-1, or Marine Corps TM-10510-14/1 to determine TMDE supportability. Refer to Air Force TO 33K-1-101, Army TB 43-180 or Navy NAVAIR 17-35NCE-1 and MIL-HDBK-300 to identify existing military department calibration equipment and standards. If existing applicable military department calibration equipment and/or standards will not meet the requirements, the design activity shall recommend equipment and/or standards and a narrative justification shall be documented.

b. The design activity shall assess specifications of the calibration equipment and standards used in support of other TMDE. The calibration equipment and standards identified shall satisfy all parameters of each measurement and comply with the requirements set forth in this standard (see 5.4.3).

c. Where several items of calibration equipment and standards are used in combination to support other TMDE, the overall test configuration specification and uncertainty shall be documented.

5.4.3 Test Uncertainty Ratio. The recommended TMDE shall be capable of measuring or generating to a higher accuracy than the measurement parameters being supported. Unless otherwise specified, a minimum TUR of 4 to 1 is desired. The actual TUR shall be documented.

a. If a TUR of 4 to 1, or the specified TUR, cannot be achieved, the design activity shall analyze the measurement requirements and provide documented justification for the lesser TUR.

b. A TUR of 4 to 1 is not required when the TMDE only provides input stimuli which are not used to characterize performance of the operational equipment or other TMDE. In this case, a minimum TUR of 1 to 1 is acceptable.

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5.4.4 Automatic Test Equipment.

5.4.4.1 ATE design. The ATE performance specifications shall be more accurate than the system or equipment operational requirements (see 5.4.3).

5.4.4.2 ATE technical description.

a. The design activity shall assess the complete measurement and stimuli capabilities that can be made available at the ATE Unit Under Test (UUT) interface.

b. The design activity shall determine the subset of ATE capabilities actually used for UUT testing.

c. TMDE recommended for test, measurement, and calibration of ATE shall be selected to support ATE capabilities actually being used for UUT testing and ATE self testing.

5.4.4.3 ATE calibration. The design activity shall assess all functions and parameters required for UUT testing, ATE self test, and calibration of the ATE. ATE calibration is implemented as a test program set (TPS) with the program running on the ATE host computer.

a. The ATE/UUT parameters shall be documented to the ATE/UUT interface.

b. The ATE calibration TPS shall be structured to provide traceability of every calibrated function and parameter to approved national or international standards. To this end, external standards and calibrated ATE components, used as working or secondary standards, shall be used.

c. The ATE calibration TPS shall calibrate the ATE to a TUR of 4 to 1 when possible (see 5.4.3). This includes, but is not limited to, the most stringent support requirements of the UUT, self-test, and the calibration TPS itself.

d. In the case of ATE that uses built-in calibration standards, these standards shall be identified with their full measurement and stimuli capabilities (see 4.1c).

e. Calibration standards or calibration procedures used to support built-in standards shall be recommended.

6. NOTES

6.1 Intended use. This document provides a means for calibration and measurement traceability to ensure system and equipment operational integrity and accuracy.

6.2 Issue of DoDISS. When this standard is used in acquisition, the applicable issue of the DoDISS will be cited in the solicitation (see 2.2.1).

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6.3 Associated Data Item Descriptions (DIDs). This standard is cited in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), as the source document for the following DID. When it is necessary to obtain the data, the applicable DID must be listed on the Contract Data Requirements List (DD Form 1423), except where the DoD Federal Acquisition Regulation Supplement exempts the requirement for a DD Form 1423.

DID Number	DID Title
DI-QCIC-80278	Calibration and Measurement Requirements Summary (CMRS)

The above DID was current as of the date of this standard. The current issue of the AMSDL must be researched to ensure that only current and approved DIDs are cited on the DD Form 1423.

6.4 Subject term (keyword) listing.

Metrology
Parameters
Specifications
Standards
Test uncertainty ratio
Tests
Traceability
Uncertainty

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - OS
Air Force – 84

Preparing Activity:
Air Force – 84

Agent:
Air Force – 99

(Project QCIC 0151)

Review Activities:
Army - AR, AT, AV, CR, TM
Navy - AS, EC, MC, NM, SH, TD
Air Force - 11, 13, 19

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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD-1839C	2. DOCUMENT DATE *(YYYYMMDD)* 20001127

3. DOCUMENT TITLE

CALIBRATION AND MEASUREMENT REQUIREMENTS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include ZIP Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME WR-ALC/TILCC ATTN: R. YOHE	b. TELEPHONE *(Include Area Code)* (1) Commercial 912-926-1183 (2) DSN 468-1183
c. ADDRESS *(Include ZIP Code)* 420 Second Street. Suite 100 Robins AFB, GA 3i098-1640	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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